

# FINAL EXPLANATION: MATHEMATICS GRADE 10

Teacher Key: Exemplar Answers and Marking Guide

## FINAL EXPLANATION: MATHEMATICS GRADE 10 – 3.2 ROTATION: FINAL EXPLANATION

Teacher Key: Exemplar Answers (do not give to students)

### INSTRUCTIONS GIVEN TO STUDENTS

This is your Final Explanation for the unit. Think back to the Ngong Hills Wind Farm video and everything you found out in Lessons 1 to 10. Answer each section in the space under it. Write in full sentences, use the correct rotation words (centre of rotation, angle of rotation, direction, object, image, order of rotational symmetry, congruent) and show any working or constructions you use. Always connect your answers to the Ngong Hills turbine. Positive angles mean anti-clockwise and negative angles mean clockwise.

### Section 1: What is a rotation and what stays the same?

**A friend watches the Ngong Hills turbine blades spinning and says, 'The blades move so much, so how can the turbine always look the same?'**

**(a) Name the three things you need to state to describe a rotation completely, and say which part of the turbine is the centre of rotation.**

**(b) State TWO properties of rotation that stay the same when the blades turn, and say what changes.**

**(c) Explain how a positive angle and a negative angle differ.**

(a) A rotation is described by a centre of rotation, an angle of rotation and a direction. On the turbine the centre of rotation is the hub, which is the fixed point and never moves.

(b) First, the distance from the centre to any point on the object equals the distance from the centre to its image ( $OA = OA'$ ), so each blade tip stays the same distance from the hub. Second, the object and its image are congruent: the shape and size do not change. Only the position of the blades changes.

(c) A positive angle means an anti-clockwise rotation and a negative angle means a clockwise rotation. For example,  $+120^\circ$  and  $-120^\circ$  move the blade in opposite directions.

### Section 2: Predicting where a point lands

**Treat the hub of the turbine as the origin  $O(0,0)$ . A blade tip is at  $P(4,2)$ .**

**(a) Find the coordinates of the image of P after a rotation of**

(a) The rule for  $90^\circ$  anti-clockwise is  $(x, y) \rightarrow (-y, x)$ , so  $P(4,2) \rightarrow P'(-2, 4)$ .

(b) The rule for  $180^\circ$  is  $(x, y) \rightarrow (-x, -y)$ , so  $P(4,2) \rightarrow (-4, -2)$ .

(c) A  $270^\circ$  anti-clockwise rotation is the same as  $90^\circ$  clockwise, with rule  $(x, y) \rightarrow (y, -x)$ , so  $P(4,2) \rightarrow (2, -4)$ .

(d) After four  $90^\circ$  rotations the total turn is  $360^\circ$ , and the tip returns exactly to  $(4, 2)$ . The path goes  $(4,2) \rightarrow (-2,4) \rightarrow (-4,-2) \rightarrow (2,-4) \rightarrow (4,2)$ . This shows that we can calculate exactly where any point on a blade will land using the rotation rules, and that a full turn

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| <p>90° anti-clockwise about O.</p> <p>(b) Find the image of P after a rotation of 180° about O.</p> <p>(c) Find the image of P after a rotation of 270° anti-clockwise about O.</p> <p>(d) What happens to the blade tip after four successive 90° anti-clockwise rotations? Explain how this helps to answer the driving question.</p> | <p>brings every point back to its starting position, so the blade looks the same.</p> |
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| Section 3: Why three blades look identical                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
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| <p>(a) The three blades are equally spaced around the hub. What is the angle between neighbouring blades?</p> <p>(b) What is the order of rotational symmetry of the three-bladed turbine? List the angles within one full turn at which the assembly maps onto itself.</p> <p>(c) Write the formula connecting the minimum angle of rotation and the order, and use it for the turbine.</p> <p>(d) Explain why the turbine looks identical after each of these turns.</p> | <p>(a) The angle between neighbouring blades is <math>360^\circ \div 3 = 120^\circ</math>.</p> <p>(b) The order of rotational symmetry is 3, because the assembly maps onto itself 3 times in one full turn. The self-mappings are at 120°, 240° and 360°.</p> <p>(c) Minimum angle = <math>360^\circ \div \text{order}</math>. For the turbine this is <math>360^\circ \div 3 = 120^\circ</math>.</p> <p>(d) After a turn of 120° anti-clockwise each blade moves exactly into the position that another blade occupied before. The blades are equally spaced around the fixed hub, so the whole assembly maps onto itself and looks unchanged. The same is true after 240° and 360°.</p> |

| Section 4: Congruence and finding a hidden centre                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
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| <p>(a) Blade 2 is the image of Blade 1 under a rotation about the hub. State the angle and direction of this rotation, and explain why this proves the three blades are congruent. Use the word isometry.</p> <p>(b) You are shown only a blade in two positions (object and</p> | <p>(a) Blade 2 is the image of Blade 1 under a rotation of 120° anti-clockwise (+120°) about the hub. Rotation is an isometry, which means it preserves lengths, angle sizes and area. So the image blade has exactly the same shape and size as the object blade, and the blades are congruent.</p> <p>(b) Step 1: Choose two points on the object and join each to its image, for example PP' and QQ'. Step 2: Using a ruler and compass, construct the perpendicular bisector of PP' and of QQ'. Step 3: The point where the two bisectors meet is the centre of rotation (the hub). Step 4: At the centre, use a protractor to measure the angle from the line joining the centre to P to the line joining the centre to P'. Step 5: State the angle with its sign: positive if anti-clockwise, negative if clockwise.</p> <p>(c) Rotation keeps the distance from the centre the same, so the centre is the same distance from P as from P'. The only points equidistant from P and P' lie on the perpendicular bisector of PP'. The same is true for Q and Q', so the centre must lie on both</p> |

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| image) and the hub is hidden. Describe step by step how you would find the centre of rotation and then the angle of rotation. (c) Explain why your method for finding the centre works. | bisectors, which meet at one point. |
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| Section 5: Final Explanation                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
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| Answer the driving question in your own words: Why do the blades of the Ngong Hills wind turbine always look the same no matter how far they rotate, and how can we use the mathematics of rotation to predict exactly where any point on the blade will land? Use the terms centre of rotation, 120°, order of rotational symmetry, congruent/isometry, and positive/negative angle. Include evidence from the unit. | The blades of the Ngong Hills turbine always look the same because the three blades are equally spaced at 120° around the fixed centre of rotation at the hub. When the turbine rotates 120° anti-clockwise (a positive rotation), each blade moves into the position of the next blade, so the whole assembly maps onto itself. This is rotational symmetry of order 3, with self-mappings at 120°, 240° and 360°. Rotation is also an isometry, so every blade in its new position is congruent to its original: lengths, angles and area are preserved, and each blade tip stays the same distance from the hub. This is true whether the turn is anti-clockwise (positive angle) or clockwise (negative angle). Using the mathematics of rotation we can predict exactly where any point lands. For a rotation about the origin hub, a blade tip P(4,2) turned 90° anti-clockwise goes to (-2,4) using $(x, y) \rightarrow (-y, x)$ , and after 180° it is at (-4,-2). If the hub is hidden, perpendicular bisectors of object-image segments locate the centre, and a protractor gives the angle and direction. |

| RUBRIC                             |                                                                                                                                                  |                                                                                            |                                                                                  |
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| Criterion                          | Excellent (4)                                                                                                                                    | Proficient (3)                                                                             | Developing (1–2)                                                                 |
| Rotation vocabulary and properties | Correctly names centre, angle and direction, states distance invariance and congruence, and uses positive/negative correctly throughout.         | States most properties and terms correctly with minor slips in wording or sign convention. | Confuses terms or properties, or omits direction or what stays the same.         |
| Coordinate calculations            | Applies all rotation rules correctly to get (-2,4), (-4,-2), (2,-4) and explains the return to (4,2) after four turns.                           | Gets most images correct with a minor error or incomplete explanation of the full turn.    | Uses wrong rules or cannot find image coordinates.                               |
| Rotational symmetry reasoning      | Gives 120° spacing, order 3, angles 120°, 240°, 360°, uses $360^\circ \div \text{order}$ , and links clearly to why the turbine looks identical. | Gives order and angle correctly but the link to the phenomenon is thin.                    | Cannot determine order or angle or does not link to the turbine.                 |
| Congruence and finding the centre  | Uses isometry to justify congruence and describes the perpendicular bisector method fully with a valid reason it works.                          | Describes the method with most steps but the reason is incomplete.                         | Method is missing or incorrect, or congruence is asserted without justification. |

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| Final Explanation quality | Coherent explanation using all required terms with accurate evidence from the unit, answering both parts of the driving question. | Answers the driving question with most terms and some evidence. | Answer is vague, lacks key terms or evidence, or answers only part of the question. |
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## ATTRIBUTION AND LICENSING

**Curriculum Alignment:** Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

**Content & Pedagogy:** Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

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